

VICTORIA JUNIOR COLLEGE
JC 2 Preliminary Examinations 2009
9747 H2 BIOLOGY PAPER 1

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Friday

1430 – 1545 H

25 September 2009

1 hour 15 mins

INSTRUCTIONS

There are **forty** questions in this paper.

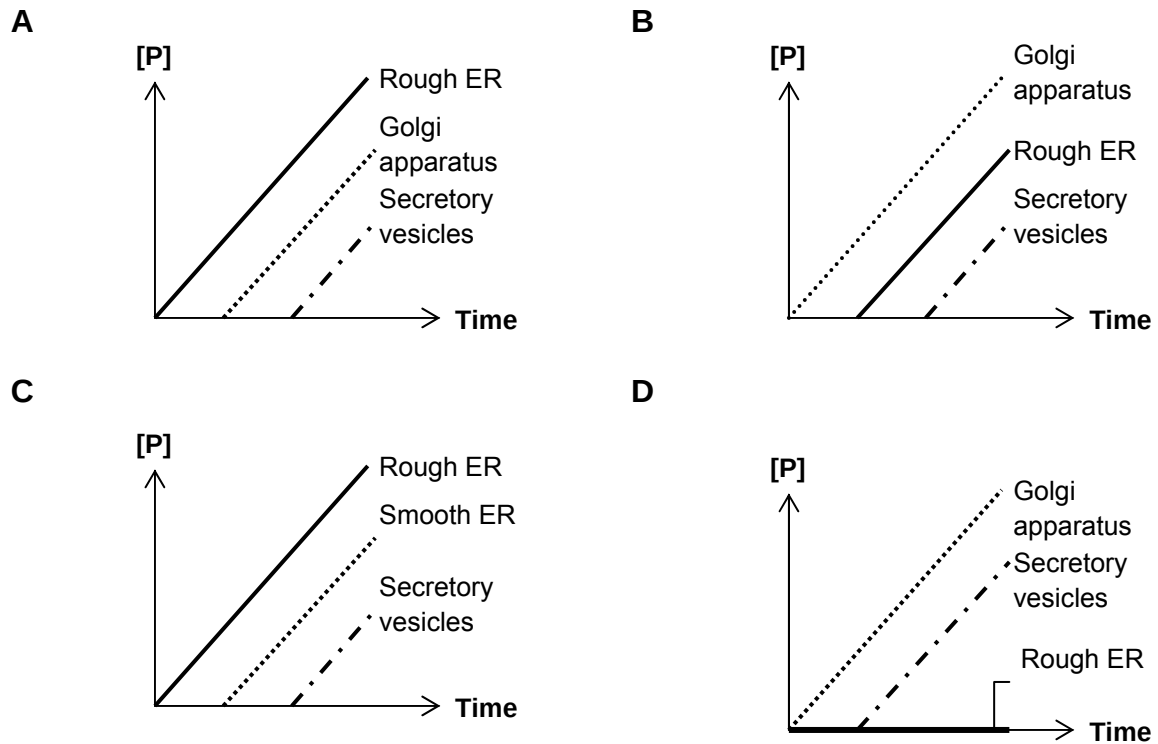
Answer **all** questions.

For each question, there are four possible options. Choose the one you consider correct and record your choice in soft pencil in a **separate** Answer Sheet.

Each correct answer will score one mark. Marks will not be deducted for wrong answers.

1 Glycoprotein P is synthesized by cells for secretion.

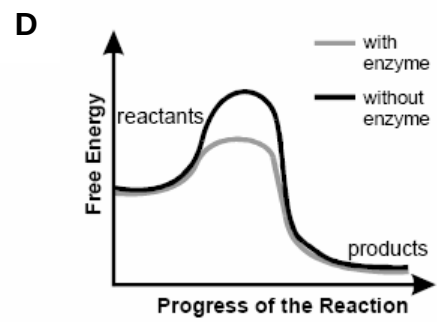
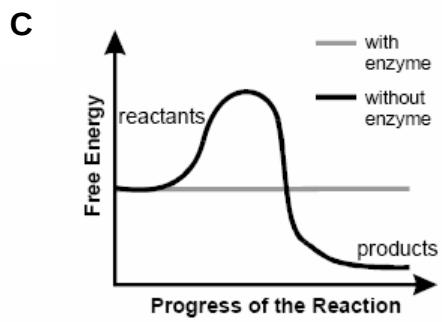
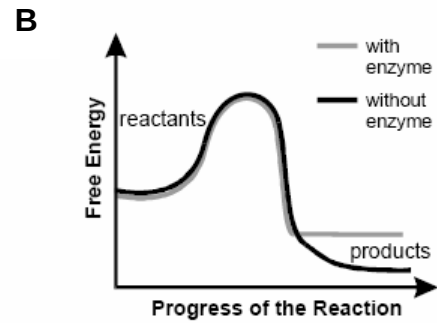
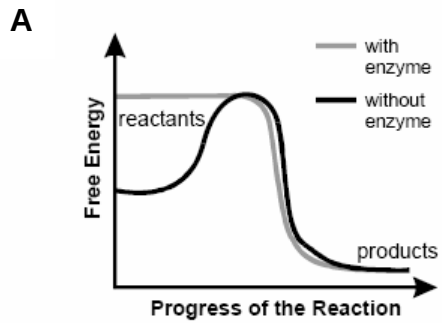
Which of the following shows the correct change in the concentration of P within the organelles after synthesis of P has begun?



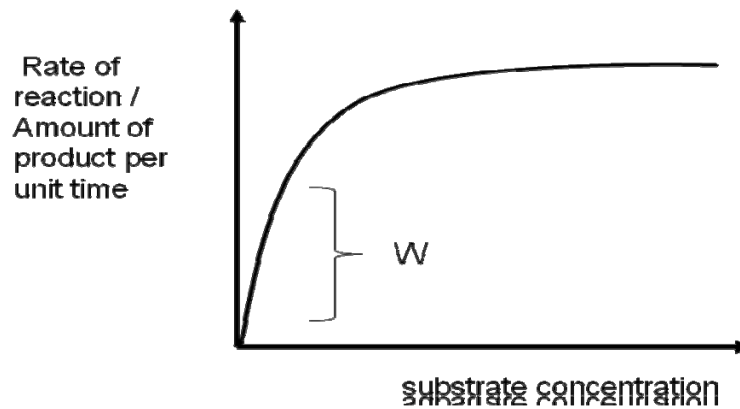
2 Which of the following best accounts for the difference in structure between amylose which is helical and cellulose which is fibrous?

- A** Presence or absence of 180° rotation of alternate subunits.
- B** Presence or absence of branching in the macromolecule
- C** Subunits are either monosaccharides or amino acids.
- D** Different tendency of macromolecule to form hydrogen bonds with water.

- 3 Which of the following graphs best illustrates how the presence of an enzyme changes the energy profile of a biochemical reaction?



- 4 The graph below shows the rate of an enzyme catalyzed reaction with increasing substrate concentration. The reaction is carried out at 37°C and a pH of 7 for all substrate concentrations.

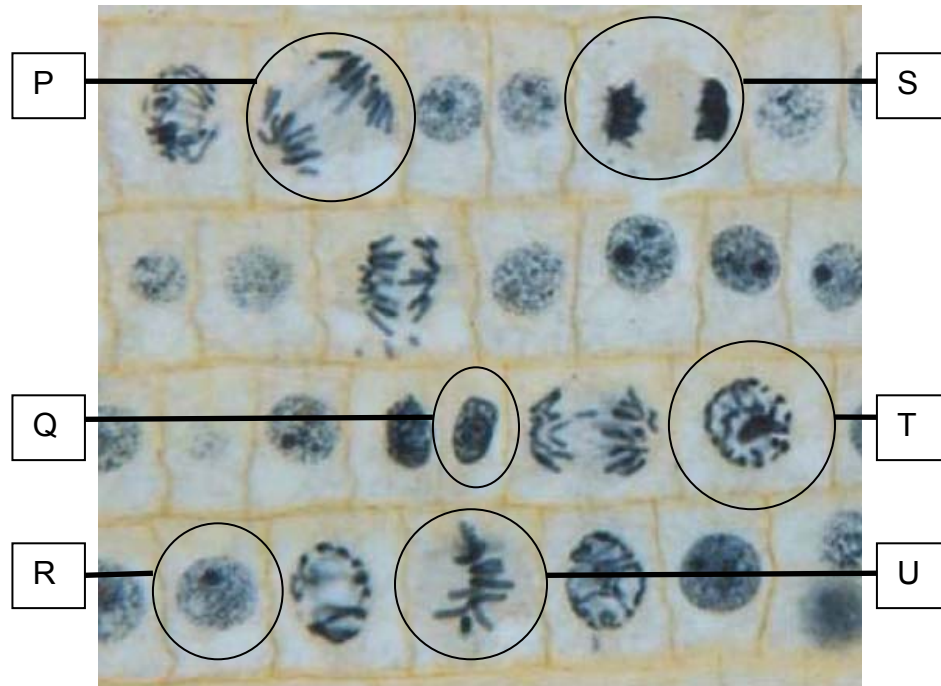


Which of the following(s) would result in a **decrease** in the rate of reaction at W?

- 1 Decrease in enzyme concentration
- 2 Decrease in temperature to 27°C
- 3 Increase in pH to 9
- 4 Addition of competitive inhibitor

- A 1 and 4 only
- B 2 and 3 only
- C 2, 3 and 4 only
- D 1, 2, 3 and 4

- 5 The photographs below show a section of the onion root tip. The cells are at different stages of mitosis.

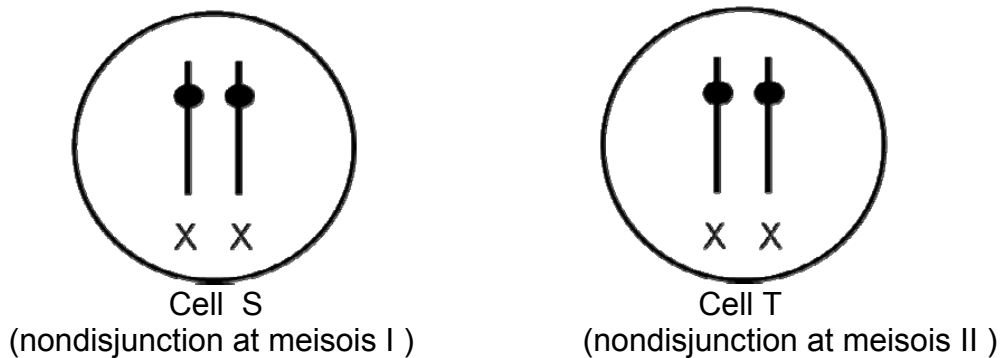


faculty.irsc.edu/.../bio%201%20resources.htm

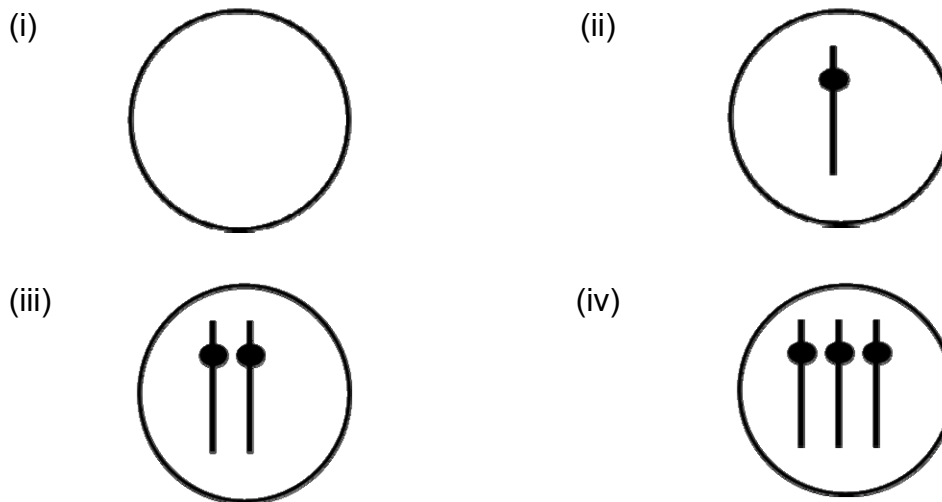
Which of the following shows the correct sequence of events that occurs in these cells?

- A R, T, U, P, S, Q
- B T, U, P, S, Q, R
- C Q, T, P, S, U, R
- D T, R, P, S, U, Q

- 6 S and T are two identical cells. Only the X chromosomes are shown in these cells. When these cells undergo meiosis to form gametes, non-disjunction of the chromosomes occurred. In cell S, it occurred during meiosis I whereas in cell T, it occurred during meiosis II.



The diagrams below show some combination of X chromosome(s) in gametes.

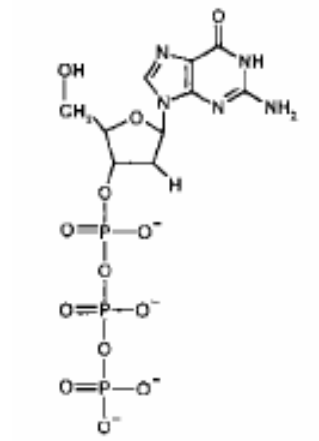


Which of the following options shows correctly the type of gametes that is/are **common** to **both** form of non-disjunction?

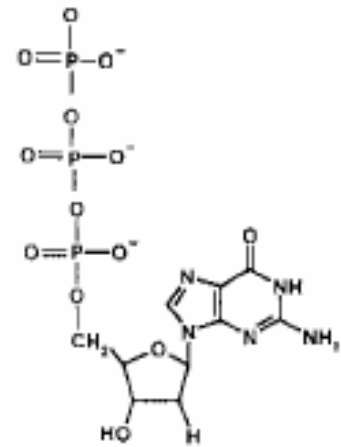
- A (i) only
- B (ii) only
- C (i) and (iii) only
- D (ii) and (iv) only

7 Which of the following is the correct physiological substrate for DNA polymerase III?

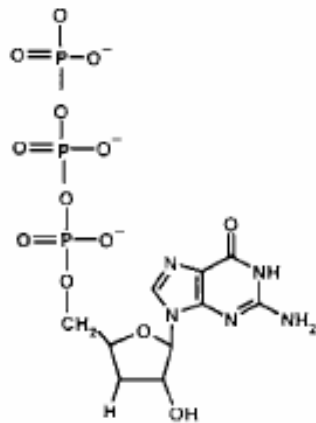
A



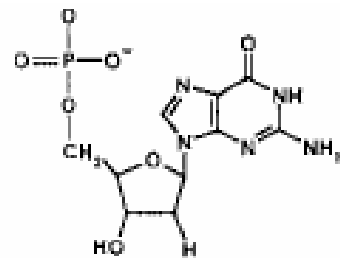
B



C



D



8

Promoter (18 nucleotides in length)	TAC	DNA sequence of 330 nucleotides	ATC
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The above diagram is a template strand of a DNA molecule. The number of amino acids in the protein coded by this template strand is

A 110

B 111

C 112

D 118

- 9 20% of the purines in a DNA molecule consist of guanine. The total percentage of thymine nucleotides in this DNA is

A 20% B 40% C 80% D cannot be determined

- 10 A DNA molecule was subjected to several gene mutations, involving the substitution of one nitrogenous base for another. The mutations have changed the base sequence of one strand of the DNA molecule. The diagram below shows the DNA molecule with one normal strand and one mutated strand.

normal A – G – C – A – T – G – G – A – T – C – C – T
mutant C – G – C – A – T – G – C – A – T – C – A – T

If both the normal and mutant DNA strands undergo 1 round of replication, what is the number of hydrogen bonds in each daughter molecule?

	<i>Number of hydrogen bonds in daughter DNA molecule made from normal DNA strand</i>	<i>Number of hydrogen bonds in daughter DNA molecule made from mutated DNA strand</i>
A	30	28
B	28	28
C	30	30
D	28	30

- 11 In the *lac* operon model for lactose metabolism in prokaryotes, what is active when the operon is turned off?

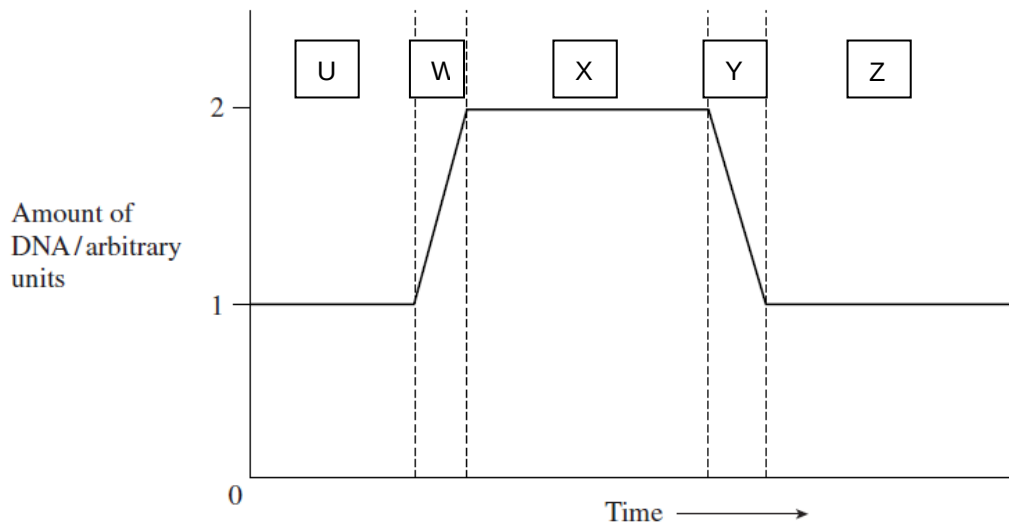
A Regulatory gene
B RNA polymerase
C Promoter region
D Structural genes

- 12 Which of the following sequences of events occurs when E coli are released from catabolite repression by transferring the bacteria to low glucose medium?
- A cAMP levels rise, cAMP binds to CAP, cAMP-CAP complex binds to DNA, activates transcription
 - B cAMP levels rise, cAMP binds to CAP, cAMP-CAP complex binds to DNA, represses transcription
 - C cAMP levels fall, cAMP is removed from CAP, CAP binds to a site on DNA and activates transcription
 - D cAMP levels fall, cAMP is removed from CAP, CAP binds to a site on DNA and represses transcription
- 13 Which one of the following is **most** likely to occur during bacterial conjugation?
- A When F^+ crosses with F^- , the recipient becomes F^+ , the donor becomes F^- .
 - B When F^+ crosses with F^- , the recipient remains F^- , the donor becomes F^+
 - C When Hfr crosses with F^- , the recipient remains F^- the donor remains Hfr
 - D When Hfr crosses with F^- , the recipient becomes Hfr, the donor remains Hfr
- 14 Which of the following statements are true of the end-replication problem of linear DNA?
1. When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
 2. Telomerase prevents the end-replication problem from occurring.
 3. Repeated rounds of replication produce shorter and shorter DNA molecules.
 4. Prokaryotes do not have the end-replication problem.
- A 1 and 2
 - B 1 and 4
 - C 2 and 3
 - D 3 and 4

- 15** Which of the following best describes the difference between tandemly repetitive and interspersed repetitive DNA?
- A** Interspersed repetitive DNA is found throughout the genome.
 - B** Tandemly repetitive DNA is composed of transposons.
 - C** Interspersed repetitive DNA is at the telomeres.
 - D** The repeated unit of tandemly repetitive DNA is longer.
- 16** Which of the following statements about the eukaryotic control elements are correct?
- (i) Attachment of the RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors.
 - (ii) Upstream regulatory elements (UREs) increase the basal activity of promoter.
 - (iii) A given gene may have multiple enhancers, each active at a different time or in a different cell type or location in the organism.
 - (iv) Repressors bind to silencer regions of DNA far upstream of promoters to repress transcription.
- A** (i) and (iv)
 - B** (ii) and (iii)
 - C** (ii), (iii) and (iv)
 - D** (i), (ii), (iii) and (iv)
- 17** In eukaryotes, many genes may have to interact with each other, requiring more regulatory proteins that can fit around a single promoter. This physical limitation is overcome by
- A** The use of very long promoters.
 - B** Alternating promoters and operators.
 - C** Distant sites in a chromosome controlling transcription of a gene.

D Having factors on one chromosome control genes on another gene

- 18** The graph shows changes in the amount of DNA in a cell during one cell cycle. The letters U – Z marks out the different phases in the cell cycle.



Many drugs that are used to treat cancer work at different time periods during the cell cycle.

- (i) Cisplatin binds to DNA and stops free DNA nucleotides from joining together.
- (ii) Drug B stops spindle fibres from shortening.

With reference to the cell cycle above, determine where these 2 drugs work.

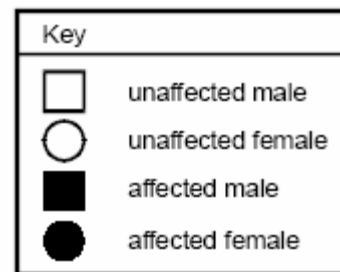
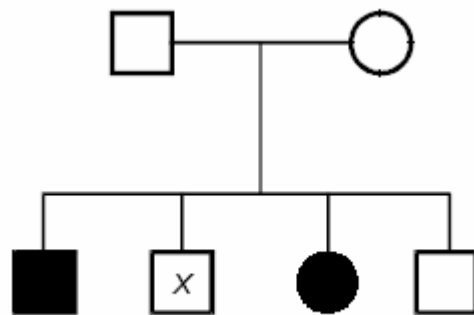
	<i>Cisplatin</i>	<i>Drug B</i>
A	W	X
B	W	Y
C	U	X
D	U	Z

- 19 Two true breeding *Drosophila* are crossed: a normal- winged, red-eyed female and a miniature-winged, vermilion-eyed male. The F1 all have normal wings and red eyes. F1 offspring are crossed with miniature-winged, vermilion-eyed flies. The following offspring were obtained:

233 normal wing, red eye
247 miniature wing, vermilion eye
7 normal wing, vermilion eye
13 miniature wing, red eye

From these data, one could conclude that the alleles for miniature wings and vermilion eye are

- A both X-linked and dominant.
B located on autosomes and dominant.
C recessive, and these genes are located 4 map units apart.
D recessive, and the deviation from the expected 9:3:3:1 ratio is due to epistasis.
- 20 The family pedigree above shows the inheritance of a trait that is determined by two alleles. What is the probability that the individual labeled X is homozygous?



- A 25%
B 33%
C 50%

D 67%

21 Two linked genes in humans have the following alleles.

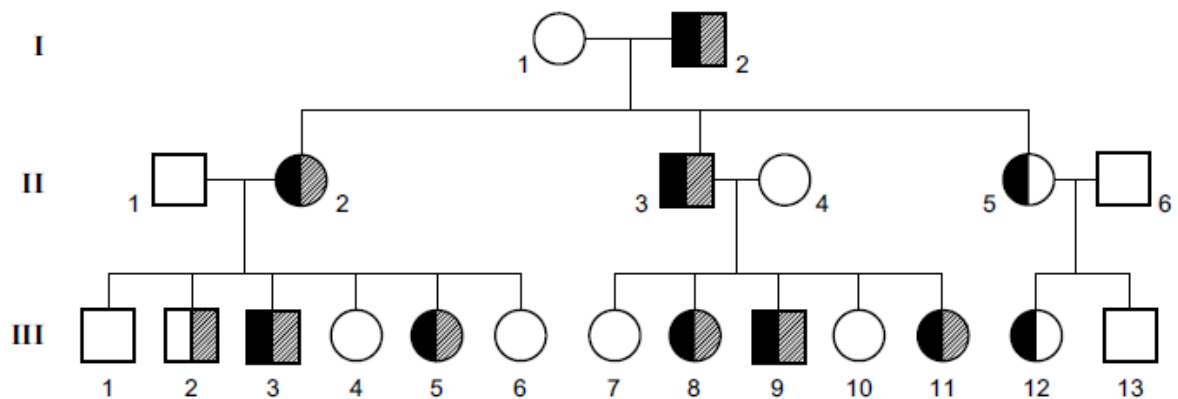
Gene 1 **A** : secretor **a** : non-secretor

Gene 2 **B** : production of protein X **b** : no protein X produced

Information regarding these traits in a family was collected and the chromosome composition of each member for these two genes was established. It was found that, in all, there were four different chromosome compositions in the family. The four types are shown in the following table.

	Type one	Type two	Type three	Type four
Genotype	$\begin{array}{c} A \\ B \end{array}$  $\begin{array}{c} a \\ b \end{array}$	$\begin{array}{c} A \\ b \end{array}$  $\begin{array}{c} a \\ b \end{array}$	$\begin{array}{c} a \\ B \end{array}$  $\begin{array}{c} a \\ b \end{array}$	$\begin{array}{c} a \\ b \end{array}$  $\begin{array}{c} a \\ b \end{array}$

The pedigree for the family is shown below.



The code for interpreting the symbols is as follows:

secretors and produce protein X 	non-secretors and produce protein X 	secretors and no protein X 	non-secretors and no protein X 
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It is reasonable to conclude that the genotype of individual

- A I-1 is type 1
- B II-2 is type 2
- C III-2 is type 3
- D III-12 is type 4

22 Assuming that all generation III individuals marry partners that are homozygous recessive for these two genes, it is reasonable to conclude that individuals

- A 5 and 6 have an equal chance of having a non-secretor child.
- B 6 and 12 have an equal chance of having a non-secretor child.
- C 5 and 6 have an equal chance of having a child who fails to produce protein X.
- D 6 and 12 have an equal chance of having a child who fails to produce protein X.

- 23** In corn, the colour of the stem can be either wild type, which is dark green, or virescent, which is pale green.

Cross	Phenotype of parents	Temperature at which raised	Offspring
1	virescent X virescent	20°C	virescent
2	virescent X virescent	37°C	wild type
3	wild type X wild type	20°C	3 wild type: 1virescent
4	wild type X wild type	37°C	wild type

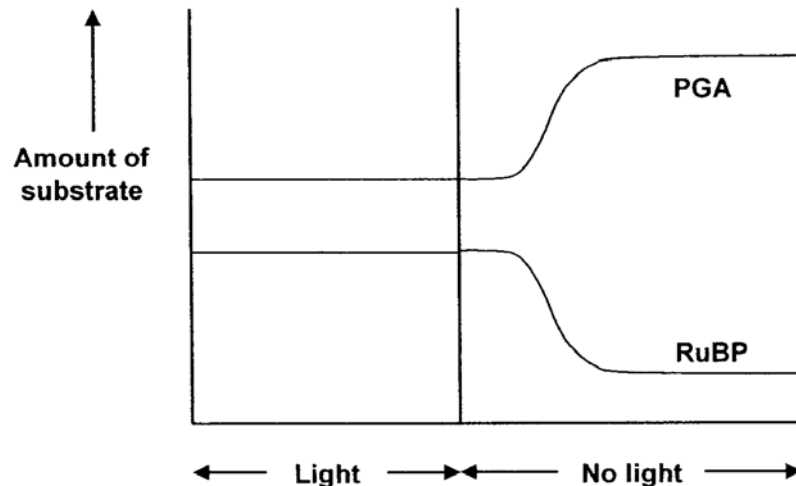
The parents of crosses 1 and 2 had the same genotypes. The parents of crosses 3 and 4 had the same genotypes.

Which of the following can be deduced from the data?

- (i) Wild type is dominant.
- (ii) The wild type parents are heterozygotes.
- (iii) Virescent is the recessive phenotype but the characteristic will only be expressed at lower temperature.
- (iv) The parents of cross 1 are only recessive at 20°C.

- A** (ii) only
- B** (i) and (iii) only
- C** (i), (ii) and (iii) only
- D** (i), (ii), (iii) and (iv)

- 24 The graph below shows the change in the concentration of RuBP and PGA in the presence of light and when the light is being turned off.



Which of following statements is **false**?

- A The amount of PGA increases due to the lack of photochemical products.
 - B In darkness, the amount of PGA increases as more RUBP is being converted into PGA.
 - C The amount of RuBP decreases due to the removal of CO_2 .
 - D In light, a steady state exists between RuBP and PGA as the RuBP that is used to from PGA is regenerated.
- 25 What is produced in cyclic photophosphorylation compared with non-cyclic photophosphorylation, using the same amount of light?
- A More ATP than in non-cyclic photophosphorylation, but no $\text{NADPH} + \text{H}^+$
 - B Less ATP than in non-cyclic photophosphorylation, but no $\text{NADPH} + \text{H}^+$
 - C Less ATP than in non-cyclic photophosphorylation, but more $\text{NADPH} + \text{H}^+$
 - D The same amount of ATP and $\text{NADPH} + \text{H}^+$ as in non-cyclic photophosphorylation.

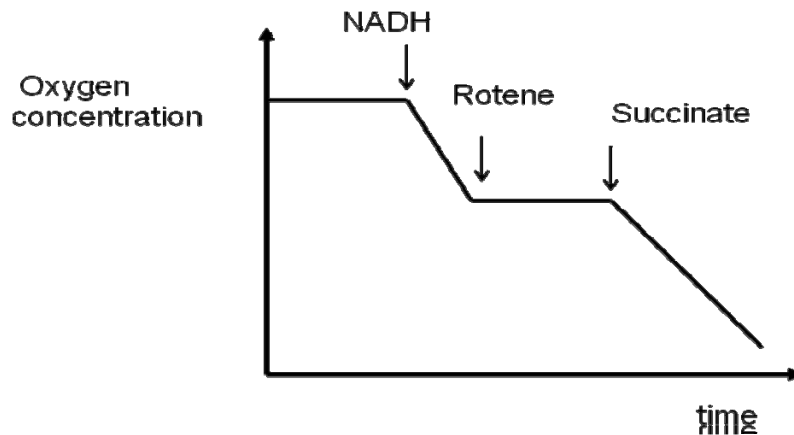
- 26 Which of the following shows correctly the differences in the products of glycolysis and Krebs cycle per glucose molecule?

	<i>Glycolysis</i>	<i>Krebs cycle</i>
A	4ATP, 1 reduced NAD	4 reduced NAD, 1 reduced FAD, 1 ATP
B	4ATP, 2 reduced NAD	8 reduced NAD, 2 reduced FAD, 2 ATP
C	2 net ATP, 1 reduced NAD	3 reduced NAD, 1 reduced FAD, 1 ATP
D	2 net ATP, 2 reduced NAD	6 reduced NAD, 2 reduced FAD, 2 ATP

- 27 Cyanide is a poison that blocks the passage of electrons along the electron transport chain. Which of the following is an intermediate metabolic effect of this poison?

- A** The pH of the intermembrane space becomes much lower than normal.
- B** Electrons are passed directly to oxygen, causing cells to explode.
- C** NADH supplies would be exhausted and ATP synthesis would cease.
- D** Proton gradient would be abolished and ATP synthesis would cease.

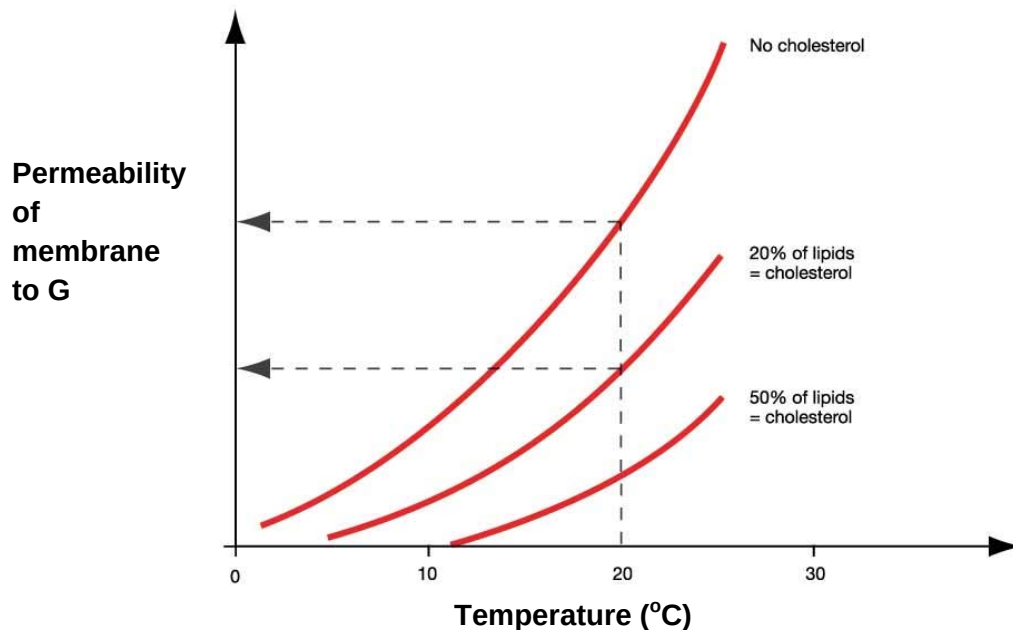
- 28 A suspension of mitochondria was prepared in a buffer containing ADP and inorganic phosphate (Pi). The oxygen concentration in the buffer was monitored carefully and recorded as shown below. At the times indicated, a specific reagent was added to the buffer. Throughout the experiment, the concentrations of ADP and Pi were in excess.



Which one of the following shows correctly from the highest to the lowest, the rate of ATP production after the addition of the three chemicals?

	<i>Highest ATP production</i>	—————→	<i>Lowest ATP production</i>
A	NADH	> Succinate	> Rotene
B	Succinate	> NADH	> Rotene
C	Rotene	> NADH	= Succinate
D	Rotene	> Succinate	> NADH

- 29 The graph below shows the permeability of three different membranes to chemical G at different temperatures. These three membranes differ in the amount of cholesterol present in the phospholipid bilayers.



Which of the following is a possible explanation for the observed data?

- A Increase in temperature increases the permeability of the membrane to G as cholesterol increases the fluidity of the phospholipids.
- B At 20°C, an increase in the proportion of cholesterol in the membrane increases the permeability of membrane to G as both cholesterol and G are non-polar.
- C Increase in the proportion of cholesterol decreases the permeability of the membrane to G as cholesterol decreases the fluidity of the phospholipids.
- D With an increase in the proportion of cholesterol in the membrane, a lower temperature is required to achieve the same level of permeability for G as G will gain a higher kinetic energy to penetrate the membrane.

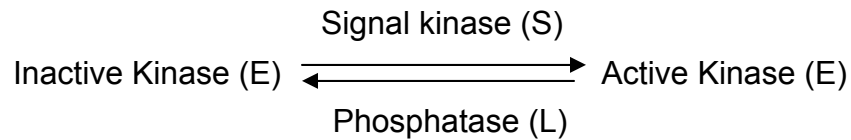
- 30** Certain drugs act at synapses and affect the action of neurotransmitter substances. The table shows the effects of four different drugs.

<i>Drug</i>	<i>Effect</i>
1	Inhibits the enzyme cholinesterase
2	Prevents the release of acetylcholine
3	Opening of Ca^{2+} gates on the pre-synaptic membrane.
4	Binds to the post-synaptic membrane receptors.

Which of the two drugs would most likely to *encourage* the response from a skeletal response to an electrical impulse arriving at the neuro-muscular junction?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

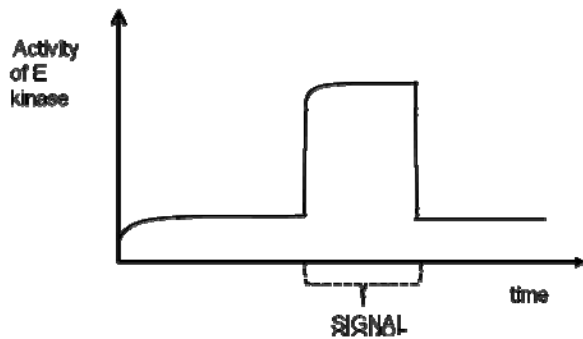
- 31 The graphs below show the computed effects of simple feedback loops. In each case, the input signal is an activated protein kinase (S) that phosphorylates and activates another protein kinase (E). E is inactivated by a protein phosphatase (L).



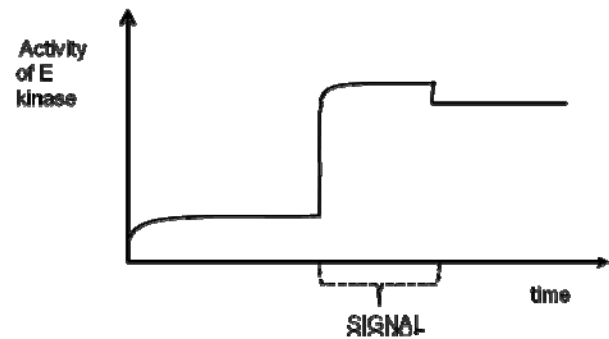
In each of the graph, the line drawn shows the activity of E over time and the dashed bracket below the x-axis indicates the time the input signal (S) is present.

Which of the graphs below shows correctly a **negative** feedback loop in the response of E to the signal?

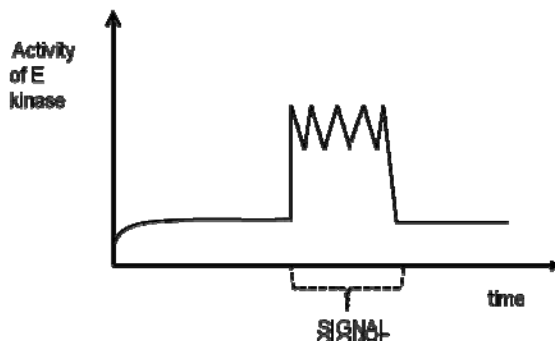
A



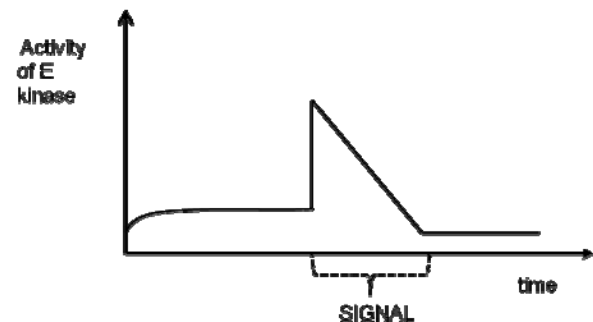
B



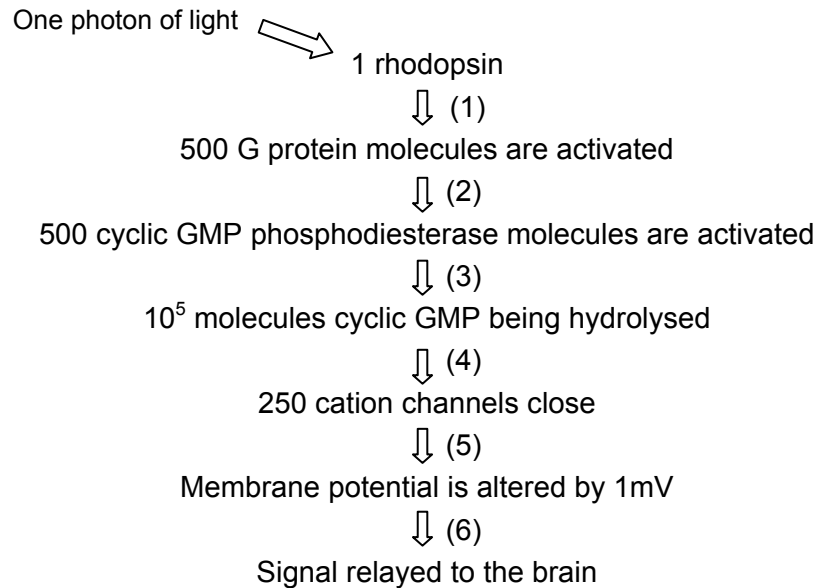
C



D



- 32** Rhodopsin, a light sensitive pigment that is present on the rods in the eyes. Rhodopsin is a G protein coupled receptor. The flow chart below shows how the chain of reactions that occur when rhodopsin absorbs a photon of light.



Identify the stages where *amplification* and *cellular responses* have occurred.

	<i>Amplification</i>	<i>Cellular response</i>
A	1, 2, 3, 4, 5	6
B	1, 3, 5	2, 6
C	1, 3	5
D	2, 3	5

- 33** A single clone of *E. coli* was cultured at 37°C for 2000 generations. A single clone was then extracted from this population and divided into replicates that were then cultured at either 32 °C, 37 °C, or 42 °C for a total of another 2000 generations. Adaptation of the new lines to 32 °C and 42 °C were periodically measured by comparing these selection lines against the ancestor population. By the end of the experiment, the lines cultured at 32 °C were shown to be 10% fitter than the ancestor population (at 32 °C), and the line cultured at 42°C was shown to be 20% more fit than the ancestor population (at 42 °C). The replicate line that was cultured at 37°C showed little improvement over the ancestral line.

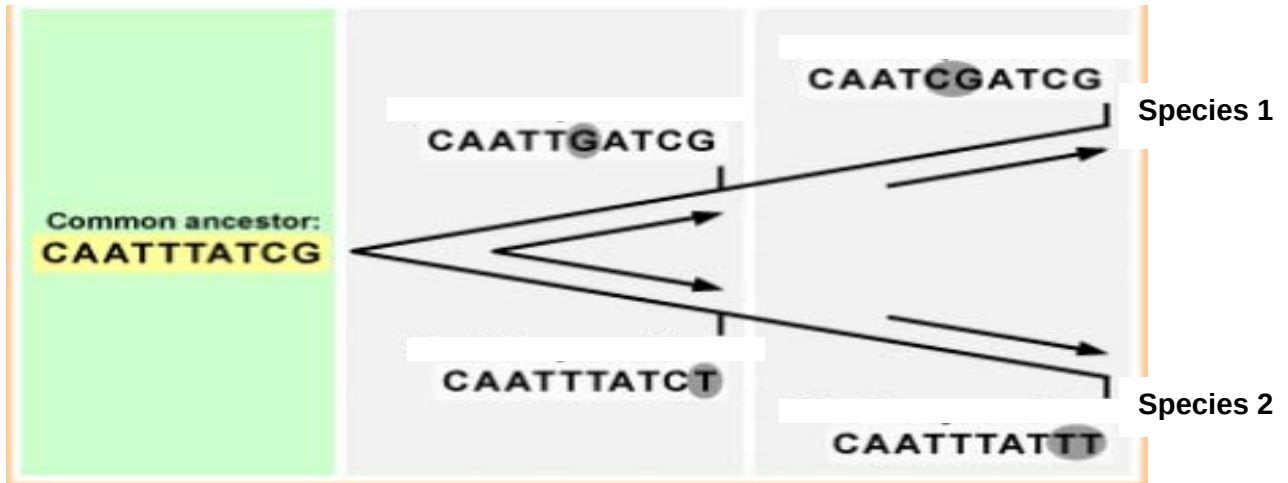
From the information, which of the following is the *best* conclusion?

- A** There are three species of *E. coli* after 4000 generations.
 - B** In the ancestral line, there exist only 3 variations of *E. coli*, i.e. *E. coli* with tolerance to 32 °C, 37 °C, or 42 °C.
 - C** The selection agent selects for the genes that enable *E. coli* to better survive in differing temperatures.
 - D** The selection agent selects for phenotypic traits that enable *E. coli* to show adaptations to high and low temperatures.
- 34** The followings are possible events that may lead to speciation.
- 1. Gene mutations
 - 2. Increased gene flow
 - 3. Natural selection
 - 4. Geographical isolation (e.g. mountain or river)
 - 5. Habitat differentiation within the same geographical location

The correct order of events that may lead to sympatric speciation is

- A** 4, 1, 3
- B** 5, 1, 3
- C** 4, 2, 3
- D** 5, 2, 3

- 35 A stretch of DNA does behave like a molecular clock. Imagine that a length of DNA found in two species differs by four bases and we know that this entire length of DNA changes at a rate of approximately one base per 25 million years. Since each lineage experienced its own evolution, the two species must have descended from a common ancestor that lived at least



(<http://evolution.berkeley.edu/evosite/evo101/IIIE1cMolecularclocks.shtml#>)

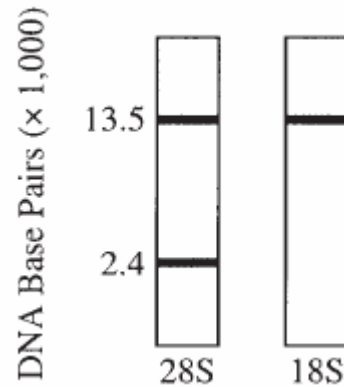
- A 25 million years ago.
 B 50 million years ago.
 C 75 million years ago.
 D 100 million years ago.
- 36 Which of the following sets of primers can be used in the PCR for the amplification of the following DNA sequence?

The dashed lines in the template sequence stand for a long sequence of bases that is to be amplified.

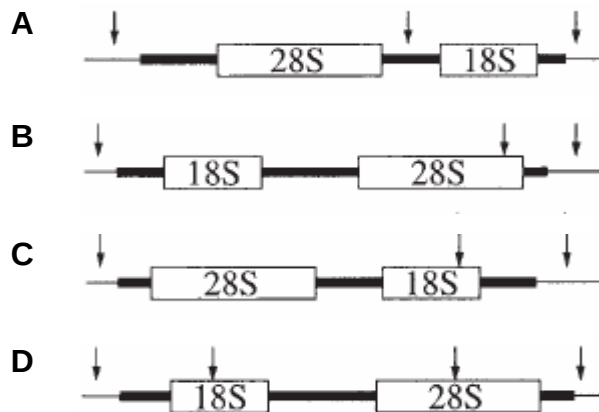
Template
 5' ATTCGGACTTG ----- GTCCAGCTAGAGG 3'
 3' TAAGCCTGAAC ----- CAGGTCGATCTCC 5'

- A 5' GTCCAGC 3' & 5' CCTGAAC 3'
 B 5' ATTCGGA 3' & 3' GATCTCC 5'
 C 5' GGACTTG 3' & 5' GCTGGAC 3'
 D 5' AUUCGGA 3' & 3' GAUCUCC 5'

- 37** The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.

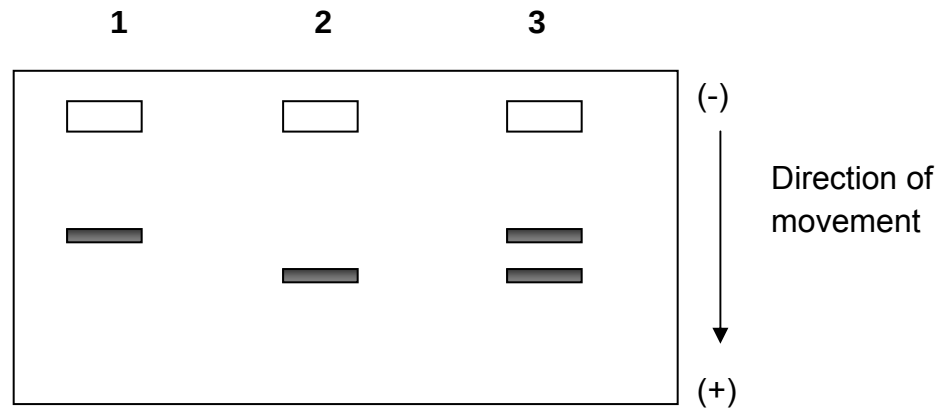


If the arrows in the following map show the location of the restriction sites of this restriction enzyme, which map *best* explains the results shown above?



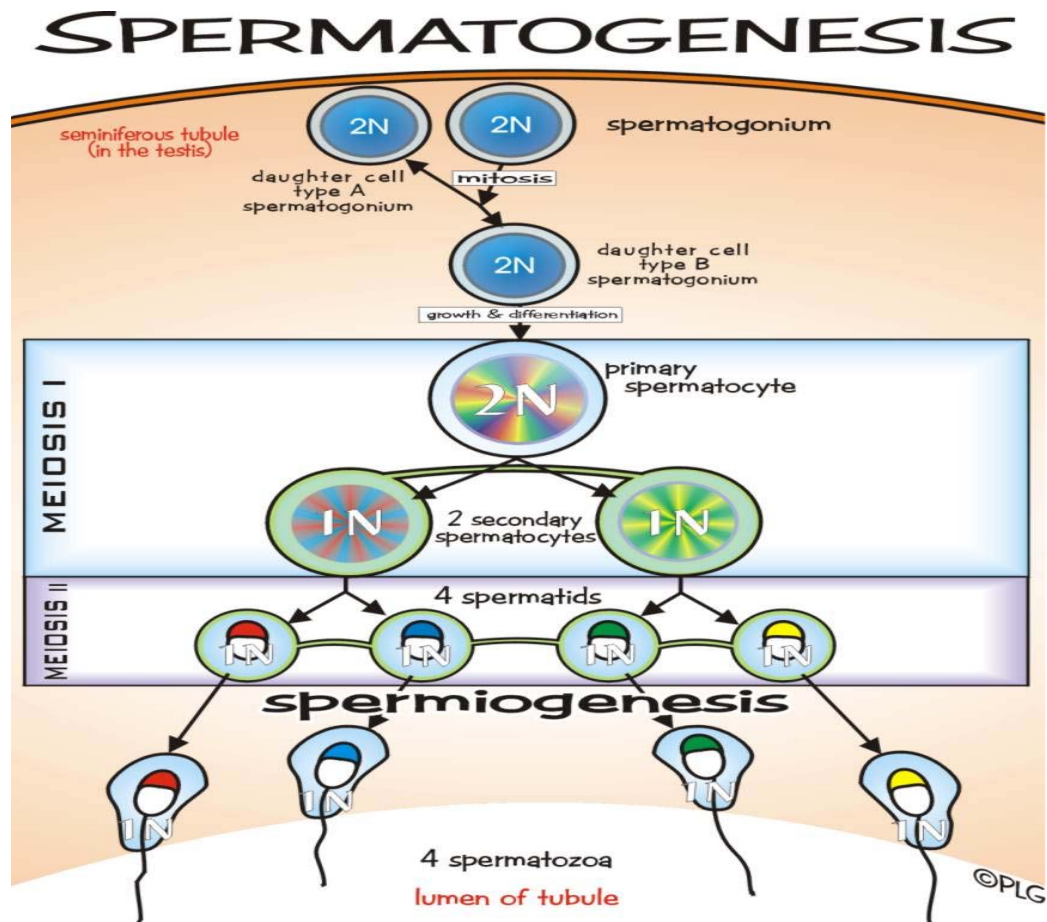
- 38** The β -globin gene can exist in two different alleles termed HbA (the normal allele) and HbS (the allele that causes sickle cell anemia in homozygotes). The polypeptides that are coded for by these two alleles differ by one amino acid. Blood samples from 3 individuals were obtained and the **proteins** separated by gel electrophoresis.

Using the gel pattern below, determine the genotypes of individuals 1 and 2 and the reason for your identification.



	Genotype of 1	Genotype of 2	Reason for identification
A	Hb ^A Hb ^A	Hb ^S Hb ^S	Since glutamic acid in the normal β -globin is negatively charged, it will move faster towards the opposite pole.
B	Hb ^S Hb ^S	Hb ^A Hb ^A	Since valine in the normal β -globin is negatively charged, it will move faster towards the opposite pole.
C	Hb ^A Hb ^A	Hb ^S Hb ^S	Since glutamic acid in the β -globin that causes sickle cell is negatively charged, it will move slower towards the opposite pole.
D	Hb ^S Hb ^S	Hb ^A Hb ^A	Since valine in the β -globin that causes sickle cell is neutral, it will move slower towards the opposite pole.

- 39 The spermatogonium is the diploid precursor of haploid sperm in the developmental pathway of mouse gametes as shown below.



(<http://science.tjc.edu/images/reproduction/spermatogenesis.jpg>)

Scientists were interested to find out which stages of sperm development have stem cell capacity. Two experiments were carried out using sterile host males with testes that lacked germ cells. The results are shown in the table below.

	Type of cells used	Result
Experiment 1	Secondary spermatocytes were injected	Fertility was not restored
Experiment 2	Spermatogonia were injected	Fertility was restored for the rest of the mouse's life

What property of a stem cell is missing in the secondary spermatocytes that results in the state of fertility not restored?

- A** Plasticity
- B** Specialization
- C** Self-renewal
- D** Transdifferentiation

40 A patient underwent gene therapy using an adenoviral mediated delivery system. The gene therapy failed and you were asked to find out the cause of the failure. Which of the following could not be the explanation behind the failure in the gene therapy?

- A** Failure of the expressed protein to be folded into the correct conformation.
- B** The CFTR protein was not expressed in adequate amount.
- C** Integration of the target gene in the enhancer region.
- D** Rejection of the vector by the host immune system.